Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 June 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

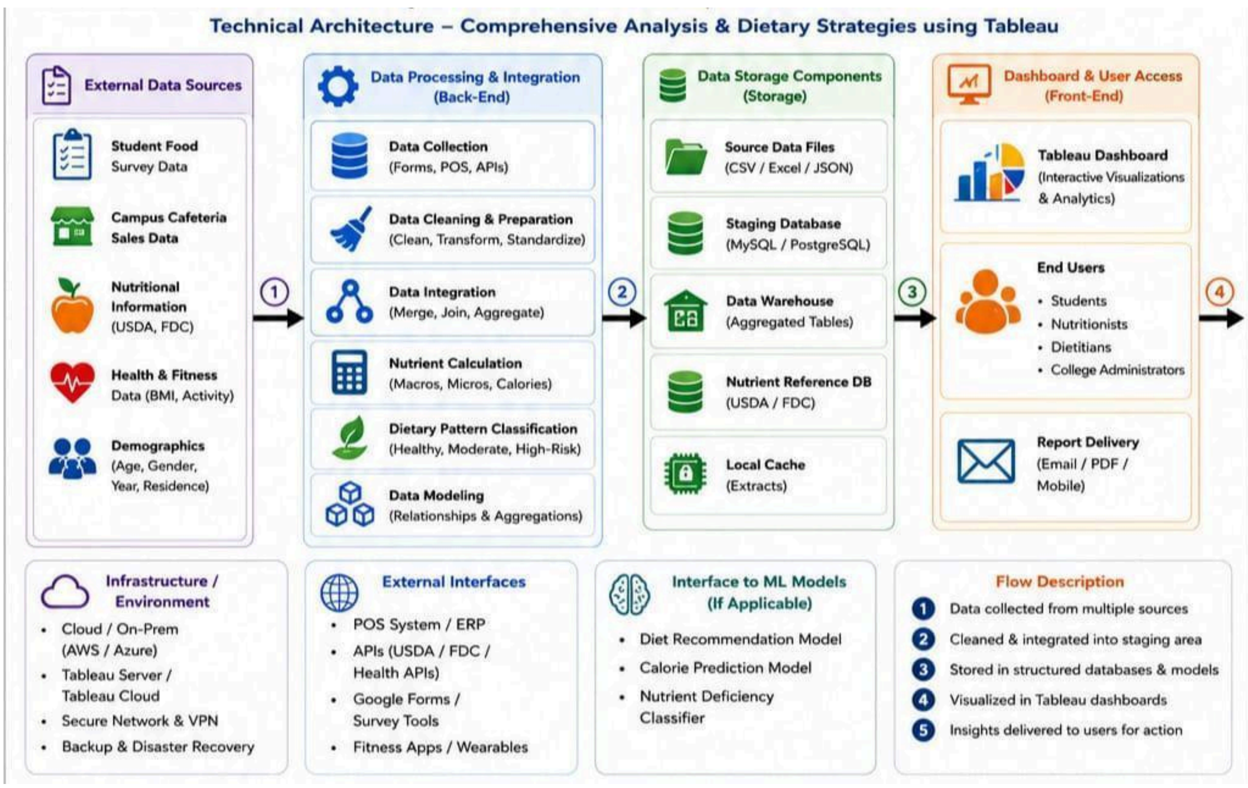

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	Food survey responses, cafeteria sales, nutrition databases, health & demographic data.	Google Forms / POS / USDA API / CSV / Excel.
2.	Data Preparation	Cleaning, transforming and loading data for analysis.	Python (Pandas) / Tableau Prep / SQL.
3.	Data Visualization Tool	Creating interactive dashboards and insights.	Tableau Desktop / Tableau Online.
4.	User Interface	Web-based dashboards for different user roles.	Tableau Public / Embedded Tableau.
5.	Access Control	Manage logins and role-based access.	Tableau Server / Active Directory.
6.	Report Sharing	Share dashboards and reports with stakeholders.	Tableau Server / PDF / Email / Mobile.
7.	Feedback Collection	Collect user feedback for dietary improvements.	Google Forms / Microsoft Forms.

Table-2: Application Characteristics:

S.NO	Characteristics	Description	Technology
1.	Usability	User-friendly dashboards with filters, tooltips and clear KPIs.	Tableau UX/UI best practices.
2.	Performance	Fast data load and responsive interactions.	Tableau Extracts (TDE/Hyper) / Optimized SQL.
3.	Security	Role-based access and secure handling of health data.	Tableau Server Permissions / HTTPS / Encryption.
4.	Scalability	Handle large datasets and increase the user base.	Tableau Server / Cloud Storage.
5.	Availability	Dashboard accessible 24/7 with minimal downtime.	Cloud Hosting (AWS/Azure) / Tableau Server.
6.	Integration	Connect with multiple data sources and external systems.	Tableau Connectors / APIs.
7.	Real-time Data Refresh	Keep dashboards updated with latest food & health data.	Tableau Scheduler